

SECTION D - APPENDICES

APPENDIX A1. PAINTING SCHEDULES FOR EXTERNAL SURFACES

Appendix Table 1.1 Painting Schedules for External Surfaces

Item Description	Substrate	Service Temperature Range	Coating System Number
Submerged Structure - Equipment / Piping / Pipe Fittings / Structures	CS	-20°C ≤ T ≤ 100°C	A-1
Splash Zone Structure / Offshore Islands - Equipment / Piping / Pipe Fittings / Structures	CS	-20°C ≤ T ≤ 100°C	A-1
Marine Jetty - Splash Zone	CS	-20°C ≤ T ≤ 100°C	A-2
Atmospheric Zone for Offshore (above splash zone)	CS	-20°C ≤ T ≤ 100°C	A-26
Atmospheric Zone for Offshore (above splash zone)	SS	-20°C ≤ T ≤ 100°C	A-27
Marine Jetty - Platform / Structures / Ladders	CS	-20°C ≤ T ≤ 100°C	A-3
Marine Jetty - Immersion Zone	CS	-20°C ≤ T ≤ 100°C	A-1
From MSL to Underside of First Deck (Offshore)	CS	-20°C ≤ T ≤ 100°C	A-2
Conductor Casing (Outer and Inner) - Submerged / Buried / Splash Zone	CS	-20°C ≤ T ≤ 100°C	A-4
		-20°C ≤ T ≤ 100°C	A-5
All Buried Steel - Piping / Equipment	CS	-20°C ≤ T ≤ 100°C	A-6
All Buried Steel - Piping / Equipment	SS / CRA	T ≤ 200°C	A-25
All Buried Steel - Tanks	CS	-20°C ≤ T ≤ 100°C	A-7
Anti-Fouling System - Submerged Zone of Offshore Structures	CS	-20°C ≤ T ≤ 100°C	A-8
Crane Deck Floor / Working Floors / Solid Deck Floors on Super Complex (Offshore)	CS	-20°C ≤ T ≤ 100°C	A-9
Solid Deck Floors on WHTs (Offshore)	CS	-20°C ≤ T ≤ 100°C	A-11
Helideck - Steel	CS	-20°C ≤ T ≤ 100°C	A-10
Helideck - Aluminium	AL	-20°C ≤ T ≤ 100°C	A-10
Carbon Steel Structural for Granular Sulphur Handling Services	CS	T ≤ 70°C	A-12
Structural Steel to be Fireproofed with Concrete	CS	Amb ≤ T ≤ 100°C	A-13
Uninsulated - Piping / Pipe Fittings / Structures / Equipment / Flares	CS	-20°C ≤ T ≤ 100°C	A-14
	CS	101°C ≤ T ≤ 200°C	A-15

Item Description	Substrate	Service Temperature Range	Coating System Number
	CS	201°C ≤ T ≤ 600°C	A-16
Insulated - Piping / Pipe Fittings / Structures / Equipment	CS	-45°C ≤ T ≤ 200°C	A-15
	CS	201°C ≤ T ≤ 600°C	A-16
	CS	Cyclic Service -20°C ≤ T ≤ 200°C	A-17
Uninsulated - Piping / Pipe Fittings / Structures / Equipment / Flares	SS	-160°C ≤ T ≤ 100°C	A-22
	SS	101°C ≤ T ≤ 600°C	A-23
Insulated - Piping / Pipe Fittings / Structures / Equipment	SS	-160°C ≤ T ≤ 100°C	A-24
	SS	101°C ≤ T ≤ 600°C	A-23
Galvanized Components HDG Gratings are generally excluded in new construction until specifically recommended by COMPANY	GS	-20°C ≤ T ≤ 100°C	A-18
Underside of Bottom Plates of Storage Tanks and Spheroids	CS	T ≤ 100°C	A-19
COS Tanks Double Shell Annulus Area (Bottom Plates and Shell 2 Metres High)	CS	T ≤ 100°C	A-20
External Shop Primer	CS	Amb ≤ T ≤ 100°C	A-21
Single Wall Storage Tanks External - Shell External - Roof Top Side - Wind Girders / Stair Tower / All CS Accessories / Structures at this Level	CS	-20°C ≤ T ≤ 100°C	A-14
Single Wall Storage Tanks Internal - Shell Internal Top 2 Metres			
Double Wall - Outer Tank Shell Internal and External / Inner Tank External and Internal Top Metres / Wind Girders / Stair Towers / All CS Accessories and Structures at this Level (Except Annulus Area Bottom Plate / Both Shells Lower 2 Metres and All CS Components at this Level)	CS	-20°C ≤ T ≤ 100°C	A-14
Double Wall - Outer Tank Shell Internal / Inner Tank Internal Top 2 Metres			
COS Tanks Double Shell Annulus Area - Bottom Plate / Both Shells Lower 2 Metres and All CS	CS	-20°C ≤ T ≤ 100°C	A-18

Notes for Appendix Table 1.1.

- For certain items (e.g. small bore "≤ 2 NPS" piping and tubing that are supplied as part of an equipment package and / or items not listed within Appendix Table 1.1). CONTRACTOR may adopt an applicable coating system previously used at COMPANY's existing facility. CONTRACTOR may propose suitable coating system for COMPANY review and approval.

APPENDIX A2. PAINTING SYSTEMS FOR EXTERNAL SURFACES

Appendix Table 2.2 Painting Systems for External Surfaces

Coating System Number	Substrate	Surface Preparation and Blast Profile	Coat	Generic System Details	DFT
A-1	CS	Sa 2.5 / SSPC-SP 10 50 μm - 75 μm	1st	Epoxy Primer (Optional, See Note 1)	50 μm
			2nd	Glass Flake Epoxy / Modified Epoxy	375 μm
			3rd	Glass Flake Epoxy / Modified Epoxy	375 μm
A-2	CS	Sa 2.5 / SSPC-SP 10 50 μm - 75 μm	1st	Epoxy Primer (Optional, See Note 1)	50 μm
			2nd	Glass Flake Epoxy / Modified Epoxy	375 μm
			3rd	Glass Flake Epoxy / Modified Epoxy	375 μm
			4th	Polyurethane	40 μm
Additional Notes for Coating Systems A-1 and A-2.					
a. New Construction: Conductor pipes shall be coated from Xmas tree top flange till sea bed. b. Site Maintenance: Conductor pipes shall be coated from Xmas tree top flange till lowest water level on the application day. c. Primer is an optional and shall be applied to facilitate mechanical works during site maintenance campaign only. Preferred method is to apply Epoxy directly on fully blasted surface on same day. d. Holiday testing shall be carried out to new construction components and for buried components. e. Total DFT of 750 microns Can be applied in a single coat application through wet on-wet technique, if required. f. Splash Zone description for UZF / SATF / UADF up to underside of the 1st deck level. g. Splash Zone description For USF / LZF / UL / NASR up to 5m above the MSL or underside of the 1st deck level.					
A-3	CS	Sa 2.5 / SSPC-SP 10 50 μm - 75 μm	1st	Inorganic Zinc Silicate	75 μm
			2nd	Epoxy MIO	100 μm
			3rd	Epoxy MIO	100 μm
			4th	Polyurethane	50 μm
A-4	CS	Sa 3 / SSPC-SP 5 50 μm - 75 μm	1st	Phenolic Novolac Epoxy	900 μm
Additional Notes for Coating System A-4.					
a. Fully cured coating shall be 100% holiday tested according to NACE RP0274 and shall be free of holidays. b. Coating shall be stripped and recoated when the number of holidays exceeded 10 in one conductor pipe. c. Detail application procedure and project specific ITP shall be submitted for COMPANY’s approval prior to commencement. d. The approved application procedure shall be qualified by PQT.					

Coating System Number	Substrate	Surface Preparation and Blast Profile	Coat	Generic System Details	DFT
e. Coated pipes shall be protected by appropriate means during storage against UV for approximate one year. f. Collar / sleeve passing test over the coated casings shall be carried to control the maximum 1100 μm DFT. g. Colour shade other than Black is acceptable. h. Outer conductor pipes shall be coated from Xmas tree top flange till sea bed.					
A-5	CS	Sa 3 / SSPC-SP 5 50 μm - 75 μm	1st	FBE Primer	350 μm
			2nd	FBE Abrasion Resistance	550 μm
			3rd	FBE Non-Slip Thermosetting Epoxy	100 μm
Additional Notes for Coating System A-5.					
a. Fully cured coating shall be 100% holiday tested according to NACE RP0274 and shall be free of holidays. b. Coating shall be stripped and recoated when the number of holidays exceeded 10 in one pipe. c. Detail application procedure and project specific ITP shall be submitted for COMPANY's approval prior to commencement. d. The approved application procedure shall be qualified by PQT. e. Coated well casings shall be protected by appropriate means during storage against ultraviolet rays for approximately one year. f. Application procedure for dual layer FBE shall be qualified by PQT. g. Collar / sleeve passing test over the coated Inner casings shall be carried to control the maximum 1100 μm DFT. h. Outer conductor pipes shall be coated from Xmas tree top flange till sea bed.					
A-6	CS	Sa 2.5 / SSPC-SP 10 50 μm - 75 μm	1st	High Build Glass Flake Epoxy / Modified Epoxy	200 μm
			2nd	High Build Glass Flake Epoxy / Modified Epoxy	200 μm
			3rd	Densopol 80 HT (or equivalent) Wrap	1.85 mm
A-7	CS	Sa 2.5 / SSPC-SP 10 50 μm - 75 μm	1st	High Build Glass Flake Epoxy / Modified Epoxy	300 μm
			2nd	High Build Glass Flake Epoxy / Modified Epoxy	300 μm
Additional Notes for Coating Systems A-6 and A-7.					
a. When the coating has fully cured, all of the coated surface shall be tested for imperfections and defects (holidays) to the NACE Standard SP-0188. b. Total DFT of 400 microns Can be applied in a single coat application through wet on-wet technique, if required. c. Apply Densopol 80 HT self-adhesive, bitumen based anti- corrosive tape that has a heavy duty PVC backing over the coated section plus 1 meter at both protruding ends to act as a backfill protective coating. Overlap each turn of wrapping tape by at least 1" (25 mm) or 55% to achieve double thickness. Ensure that entire surface is covered with no gaps and air pockets.					

Coating System Number	Substrate	Surface Preparation and Blast Profile	Coat	Generic System Details	DFT
d. Holiday testing as per NACE SP0188 shall be conducted for all of the Denso wrapped surface with a set voltage recommended by Denso Tape MANUFACTURER.					
A-8	CS	Sa 2.5 / SSPC-SP 10 50 μm - 75 μm	1st	Glass Flake Epoxy / Modified Epoxy	300 μm
			2nd	Glass Flake Epoxy / Modified Epoxy	300 μm
			3rd	Sealer Coat	75 μm - 100 μm
			4th	Anti-Fouling	75 μm - 125 μm
			5th	Anti-Fouling	75 μm - 125 μm
Additional Notes for Coating System A-8.					
a. Application shall note that some Coating MANUFACTURER's recommend that the tie-coat or the anti-foulant coat should be applied into the corrosion protection coat while tacky. Coating MANUFACTURER's recommendation in this regard shall be strictly observed.					
A-9	CS	Sa 2.5 / SSPC-SP 10 50 μm - 75 μm	1st	Surface Tolerant Epoxy	150 μm
			2nd	Anti-Skid HB Abrasion Resistant Coating	2000 μm
			3rd	Polyurethane	50 μm
A-10	CS	Sa 2.5 / SSPC-SP 10 75 μm - 100 μm	1st	Surface Tolerant Epoxy	150 μm
			2nd	Anti-Skid HB Abrasion Resistant Coating	1000 μm - 3000 μm
			3rd	Polyurethane	50 μm
	AL	Sa 1 / SSPC-SP7 25 μm - 35 μm	1st	Surface Tolerant Epoxy	150 μm
			2nd	Anti-Skid HB Abrasion Resistant Coating	1000 μm - 3000 μm
			3rd	Polyurethane	50 μm
Additional Notes for Coating System A-10.					
a. Coating MANUFACTURER / SUB-CONTRACTOR shall ensure that the applied anti-skid coating system meets the requirements of CAAP 071 and the friction values of 0.85 is achieved.					
b. In case, the friction values are lower than the desired value of 0.85, then Coating MANUFACTURER / SUB-CONTRACTOR shall repair the system to achieve the desired friction properties.					
c. Only, GCAA approved organisation shall be allowed to conduct and certify the friction values.					
d. Helideck marking / visual aids shall be in accordance to CAAP 071 requirement.					
e. Specific marking sketch to be submitted to COMPANY for review / approval prior to commencement of project.					

Coating System Number	Substrate	Surface Preparation and Blast Profile	Coat	Generic System Details	DFT
A-11	CS	Sa 2.5 / SSPC-SP 10 40 μm - 65 μm	1st	Glass Flake Epoxy	200 μm
			2nd	Glass Flake Epoxy	200 μm
			3rd	Coarse Texture Non-Spark Aggregate	0.8mm - 1.2mm
			4th	Polyurethane	60 μm
Additional Notes for Coating System A-11.					
a. Anti-skid material shall be applied during application of the 2nd coat.					
A-12	CS	Sa 2.5 / SSPC-SP 10 50 μm - 75 μm	1st	High Chemical Resistance Paint (Resistance to 98% Sulphuric Acid)	Refer Coating MANUFACTURER
A-13	CS	Sa 2.5 / SSPC-SP 10 50 μm - 75 μm	1st	Zinc Rich Primer	75 μm
			2nd	Polyamide Epoxy MIO	125 μm
			3rd	Polyurethane	75 μm
A-14	CS	Sa 2.5 / SSPC-SP 10 50 μm - 75 μm	1st	Inorganic Zinc Silicate	75 μm
			2nd	Polyamide Epoxy MIO	100 μm
			3rd	Polyamide Epoxy MIO	100 μm
			4th	Polyurethane	50 μm
Additional Notes for Coating System A-14.					
a. Prior to the application of intermediate coat, the surface of inorganic zinc silicate primers shall be sealed by application of a tie coat by mist coat / wet on wet method to seal off the surface of the zinc and to avoid solvent entrapment and post application blistering.					
b. Maximum DFT of inorganic zinc silicate shall be 90 μm to avoid any mud-cracking.					
A-15	CS	Sa 2.5 / SSPC-SP 10 50 μm - 75 μm	1st	Phenolic Epoxy	150 μm
			2nd	Phenolic Epoxy	150 μm
Additional Notes for Coating System A-15.					
a. The applied coating system shall be free of any defects and shall perform satisfactory during the entire construction stage till commissioning at ambient temperature.					
b. Perform satisfactorily during non-operative phase in CX environment.					
A-16	CS	Sa 2.5 / SSPC-SP 10 50 μm - 75 μm	1st	Inert Multi Polymeric Matrix	150 μm
			2nd	Inert Multi Polymeric Matrix	150 μm
Additional Notes for Coating System A-16.					
a. The applied coating system shall be free of any defects and shall perform satisfactory during the entire construction stage till commissioning at ambient temperature.					
b. Perform satisfactorily during non-operative phase in CX environment.					

Coating System Number	Substrate	Surface Preparation and Blast Profile	Coat	Generic System Details	DFT
A-17	CS	Sa 2.5 / SSPC-SP 10 50 μm - 85 μm	1st	Thermally Sprayed Aluminium	200 μm
	SS	SSPC-SP 16 35 μm - 50 μm	1st	Thermally Sprayed Aluminium	200 μm
A-18	CS / GS	SSPC-SP 16 35 μm - 50 μm	1st	Surface Tolerant Epoxy Primer	125 μm
			2nd	HB Epoxy	125 μm
			3rd	Polyurethane	50 μm
Additional Notes for Coating System A-18.					
a. Unless specifically mentioned by COMPANY, HDG gratings are excluded from over coating in new constructions.					
A-19	CS	Sa 2.5 / SSPC-SP 10 50 μm - 75 μm	1st	Surface Tolerant HB Glass Flake Epoxy	300 μm
			2nd	Surface Tolerant HB Glass Flake Epoxy	300 μm
Additional Notes for Coating System A-19.					
b. Coating shall only be applied in combination with effective cathodic protection and shall be consulted with COMPANY prior to application.					
A-20	CS	Sa 2.5 / SSPC-SP 10 40 μm - 65 μm	1st	Epoxy Primer (Optional, See Note 1)	50 μm
			2nd	Epoxy Glass Flake - Reinforced	150 μm
			3rd	Epoxy Glass Flake - Reinforced	150 μm
			4th	Polyurethane	50 μm
A-21	CS	Sa 2.5 / SSPC-SP 10 10 μm - 25 μm	1st	Weldable Inorganic Zinc	25 μm
A-22	SS	SSPC-SP 16 35 μm - 50 μm	1st	Phenolic Epoxy	125 μm
			2nd	Phenolic Epoxy	125 μm
			3rd	Polyurethane	40 μm
A-23	SS	SSPC-SP 16 35 μm - 50 μm	1st	Inert Multi Polymeric	125 μm
			2nd	Inert Multi Polymeric	125 μm
A-24	SS	SSPC-SP 16 35 μm - 50 μm	1st	Phenolic Epoxy	125 μm
			2nd	Phenolic Epoxy	125 μm
A-25	SS / CRA	SSPC-SP 16 35 μm - 50 μm	1st	Glass Flake Epoxy	200 μm
			2nd	Glass Flake Epoxy	200 μm

Coating System Number	Substrate	Surface Preparation and Blast Profile	Coat	Generic System Details	DFT
A-26	CS	Sa 2.5 / SSPC-SP 10 45 μm - 70 μm	1st	Inorganic Zinc Silicate	75 μm
			2nd	Polyamide Epoxy MIO	100 μm
			3rd	Polyamide Epoxy MIO	100 μm
			4th	Polyurethane	50 μm
Additional Notes for Coating System A-26.					
a. Prior to the application of intermediate coat, the surface of inorganic zinc silicate primers shall be sealed by application of a tie coat by mist coat / wet on wet method to seal off the surface of the zinc and to avoid solvent entrapment and post application blistering. b. Maximum DFT of inorganic zinc silicate shall be 90 μm to avoid any mud-cracking.					
A-27	SS	SSPC-SP 16 35 μm - 50 μm	1st	Non Inhibitive Epoxy Primer	35 μm
			2nd	HB Epoxy	100 μm
			3rd	Polyurethane	40 μm
Additional Notes for Coating System A-27.					
a. The primer shall be zinc free and non-inhibitive Epoxy type. b. The coating for stainless steel surfaces, nickel based alloys and copper based alloys and elements shall not contain any zinc and a chloride content shall be less than 25 ppm.					

Notes for Appendix Table 2.2.

1. Epoxy Primer shall be an optional, recommended practice is to apply Glass Flake Epoxy direct to blasted substrate.
2. For PQT requirements for Systems A4 and A5, see APPENDIX A3.
3. Potable water to be used for washing with high pressure of minimum 3500 psig / 241.32 barg.
4. Where applicable, the acceptance criteria for potable water used for washing shall contain maximum 50 ppm chlorides and pH value of 6.5 to 8.
5. Coating MANUFACTURER shall provide test certificate of zinc content level tested by third party laboratory. The metallic zinc content in DFT shall be at least 85% by weight and complies with the compositional requirements for SSPC Paint 20 Type II level 1, ISO 12944-5 or equivalent.
6. Coating MANUFACTURER shall provide test certificate of MIO content level tested by third party laboratory. The minimum MIO shall be 80% by weight in total pigment.
7. The minimum glass flake level shall be 15% on total pigmentation by weight. Coating MANUFACTURER shall provide third party laboratory test certificate for the glass flake level in the high build epoxy paint.